

Energy and Work

AI Tutor - Physics - Lesson 3

What is Energy?

Energy is the capacity to do work or cause change. It exists in many forms and can be converted from one form to another, but the total amount is always conserved.

- **Kinetic energy** - energy of motion.
- **Potential energy** - stored energy due to position (e.g. height).
- **Thermal energy** - energy from the motion of particles (heat).

Work

In physics, work is done when a force moves an object over a distance: $\text{Work} = \text{Force} \times \text{distance}$. Simply pushing on a wall that doesn't move does no physical work, even though it feels tiring.

Conservation of Energy

The law of conservation of energy states that energy cannot be created or destroyed, only transformed. A ball dropped from a height converts potential energy into kinetic energy as it falls.